Journal of Consciousness & Metaphysics | Research Thesis | 2026

The Architecture of Perception:

Illusions as Mismatched Geometry in the Fabric of Reality

Executive Overview, Foundational Axioms, and Mathematical Formulation of Filter Theory and Samādhi

IOTechX Epistemology Lab

Independent Research Network

https://iotechx.github.io/

Abstract. Human perception operates as a continuous decoding engine that attempts to reconstruct an underlying, undivided state of reality. The central axiom of this monograph is that *observation itself is the root cause of illusion*: any act of observation requires a bounded point of view, a division between observer and observed, and a selective filter. We formalise this claim as *Filter Theory*. Reality is modelled as a continuous field $f(x)$; an observer is modelled as a windowing operator $w(x-x_i)$ centred at a bounded location x_i; and an observation is the filtered fragment $S_i = f(x) \cdot w(x - x_i)$. Primary (Level 1) illusion is the unexamined identification $S_i \equiv f(x)$. Secondary (Level 2) illusion arises when fragments produced by non-identical filters are stitched together: the resulting boundary mismatch, the *cognitive delta* Δ, is shown to vanish for all fields if and only if the filters coincide on their overlap. We connect the framework to the Nyquist–Shannon sampling theorem, the time–frequency uncertainty principle, rate–distortion theory, the quantum measurement problem, and the classical Advaita Vedānta text *Dṛg-Dṛśya-Viveka*, and we derive a four-step deconstruction protocol whose limiting state ($w \to 1$, $\Delta \to 0$) is identified with the contemplative state of *samādhi*. Throughout, we distinguish carefully between results that are mathematical consequences of the model, analogies to established theories, and metaphysical postulates, and we close with an explicit empirical programme and a statement of limitations.

Keywords: Filter Theory; *Dṛg-Dṛśya-Viveka*; *Upādhi*; Cognitive Delta; Samādhi; Signal Processing; Advaita Vedānta.

Fields: Epistemology; Advaita Vedānta; Signal Processing; Information Theory; Philosophy of Mind.

Licence. © 2026 IOTechX Epistemology Lab. Released under the Creative Commons Attribution-NonCommercial 4.0 International licence (CC BY-NC 4.0).

Conflicts of interest. None declared. **Data availability.** All figures in this paper are generated from closed-form expressions given in the text; no empirical dataset was used.

Contents

1 Introduction **3**

1.1 Motivation . . . 3

1.2 Contributions . . . 3

1.3 Epistemic status of the claims . . . 3

1.4 Organisation . . . 4

2 Background and Related Work **4**

2.1 Filter and reduction-valve theories of mind . . . 4

2.2 Constructive perception and interface theories . . . 4

2.3 Cognitive dissonance . . . 4

2.4 Signal processing and information theory . . . 4

2.5 Quantum measurement . . . 5

2.6 Advaita Vedānta and contemplative practice . . . 5

3 Formal Framework **5**

3.1 Primitive objects . . . 5

3.2 Axioms . . . 5

3.3 Operational causal chain . . . 6

4 Primary Illusion: Observation as a Filter Function **6**

4.1 The three artificial constructs . . . 6

4.2 Mathematical formulation of perception . . . 7

4.3 Parallels in Advaita and quantum mechanics [A] . . . 7

5 Topology of Separation and Fragmented Sampling **7**

5.1 The cup and the ocean . . . 8

5.2 Three structural properties of filters . . . 8

5.3 Inequality of the sum and the whole . . . 9

6 Secondary Illusion: Mismatched Interlocks and Cognitive Delta **9**

6.1 Truth condition versus illusion condition . . . 10

6.2 The nature and phenomenology of delta . . . 10

6.3 Mechanics of forced interlocking . . . 11

6.4 Truth versus illusion: summary . . . 12

7 Signal-Processing and Information-Theoretic Analogies **12**

7.1 Nyquist–Shannon sampling . . . 12
7.2 Windowing, uncertainty and the capacity of a filter . . . 13
7.3 The mind as a lossy codec . . . 13
7.4 Summary of analogies . . . 14

8 Deconstruction: The Four-Step Protocol to Samādhi 14
8.1 The four steps . . . 14
8.2 Definition of samādhi in the model . . . 15
8.3 The equation of liberation . . . 15
8.4 Decoding pipeline and edge geometry . . . 15

9 Classical Foundations: *Dṛg-Dṛśya-Viveka* as Filter Theory 15
9.1 Core principle and equation mapping . . . 15
9.2 Three hierarchical levels of *Dṛśyam* . . . 16
9.3 The negative equation of viveka . . . 16

10 Practical *Sādhanā* and the Dynamics of Grace 17
10.1 The daily four-question inquiry . . . 17
10.2 Intended results . . . 17
10.3 Divine grace (*anugraha*) and vertigo as revelation . . . 17

11 Empirical Programme and Falsifiability 17

12 Discussion and Limitations 18
12.1 Interpretive scope . . . 18
12.2 Limitations . . . 18
12.3 Relation to prior work . . . 19

13 Conclusion 19

A Master Summary of Mathematical Formulations 20

B English Rendering of the Malayalam Synthesis 20

C Glossary of Sanskrit Terms 21

1. Introduction

1.1 Motivation

Traditional models of cognition, and the many self-help and therapeutic systems built upon them, pose the question: *How can we avoid illusion while observing?* The framework developed here inverts that question and asks a more radical one: *Is observation itself the primary illusion?*

Observation cannot occur in a vacuum. To observe anything, an observer must (i) assert a bounded location from which the observation is made, (ii) create a conceptual and spatial duality between the observer and the observed, and (iii) admit information only through a selective mechanism. Each of these is a restriction imposed on an otherwise seamless field. All secondary cognitive errors, including psychological strain, dogma, ideology and systemic delusion, are then *cascade errors* stemming from this initial act of separation.

1.2 Contributions

This paper makes the following contributions.

(C1) **A formal model.** We define reality, the observer-filter and the observed fragment in a single operator formalism (Section 3) and state the axioms on which the theory rests.

(C2) **Two precise results.** We prove (Proposition 1) that a family of filtered fragments recovers the whole if and only if the filters form a partition of unity, which finite, bounded observers cannot satisfy; and (Proposition 2) that the boundary-mismatch error Δ vanishes for all fields if and only if the two filters agree on their overlap.

(C3) **A cross-disciplinary mapping.** We relate the model to sampling theory, uncertainty principles, rate–distortion theory, quantum measurement and to the classical Advaita analysis of *dṛg* (seer) and *dṛśya* (seen) (Sections 7 and 9).

(C4) **A deconstruction protocol.** We state a four-step procedure and a daily four-question inquiry, and define *samādhi* in the model's terms (Sections 8 and 10).

(C5) **An empirical programme and an honest account of limits.** We list testable consequences and the model's self-referential and epistemic limitations (Sections 11 and 12).

1.3 Epistemic status of the claims

Because this work crosses mathematics, cognitive science and contemplative philosophy, we label each major statement by status:

- **[M] Mathematical result:** follows deductively from the stated definitions.
- **[A] Analogy:** a structural correspondence with an established theory; it is *not* a claim of identity.
- **[P] Postulate or metaphysical thesis:** assumed, not derived, and accepted or rejected on philosophical grounds.

1.4 Organisation

Section 2 reviews related work. Section 3 sets out the formal framework. Sections 4–6 develop primary and secondary illusion. Section 7 gives the signal-processing and information-theoretic analogies. Section 8 presents the deconstruction protocol. Section 9 grounds the framework in *Dṛg-Dṛśya-Viveka*. Section 10 gives the practical inquiry. Sections 11–13 cover the empirical programme, limitations and conclusions. Appendices collect the formulas, a glossary, and an English rendering of the Malayalam synthesis that accompanied the original draft.

2. Background and Related Work

2.1 Filter and reduction-valve theories of mind

The idea that the brain and senses *restrict* rather than *produce* awareness has a long lineage. Bergson (1911) argued that perception selects from a wider field only what is useful for action. James (1898) proposed that the brain transmits, and thereby limits, a larger consciousness, and Huxley (1954) popularised the notion of the brain as a "reducing valve". Filter Theory is in this tradition but differs in being stated as an explicit operator model with quantitative consequences.

2.2 Constructive perception and interface theories

In critical philosophy, Kant (1781) distinguished phenomena, as they appear under the forms of sensibility and understanding, from things in themselves. In contemporary work, Hoffman (2019) argues, using evolutionary game theory, that perceptual systems evolve to track fitness rather than veridical structure, so that perception is an interface rather than a window. Metzinger (2003) analyses the phenomenal self as a transparent model. Predictive-processing accounts (Friston, 2010) treat perception as inference under a generative model. Each of these converges on the thesis that perception is an active, model-laden reconstruction; Filter Theory adds a specific geometric account of *why reconstructions from several observers fail to match*.

2.3 Cognitive dissonance

Psychological tension arising from inconsistent cognitions was systematised by Festinger (1957). In our terminology, dissonance is the experiential correlate of the cognitive delta Δ (Section 6); the model supplies a geometric interpretation, namely that the inconsistency lies at the boundaries of fragments produced by different filters.

2.4 Signal processing and information theory

Windowing is a standard tool in harmonic analysis (Harris, 1978; Gabor, 1946). Sampling and reconstruction are governed by the Nyquist–Shannon theory (Nyquist, 1928; Shannon, 1949); the limits of simultaneous time and frequency concentration by the Slepian–Landau–Pollak theory (Slepian and Pollak, 1961); lossy compression by rate–distortion theory (Shannon, 1959; Cover and Thomas, 2006). We use these as formal analogues for the structure of cognitive filtering.

2.5 Quantum measurement

In the standard formalism, a quantum state evolves unitarily until a measurement, which requires a partition (the Heisenberg "cut") between system and apparatus (von Neumann, 1932). Decoherence explains the *appearance* of classical outcomes through environmental entanglement (Zurek, 2003). We use the measurement problem only as an analogy for the dependence of an outcome on the cut.

2.6 Advaita Vedānta and contemplative practice

Advaita Vedānta holds that the ultimate reality is non-dual consciousness (*Brahman*), and that the apparent multiplicity of subject and object arises through limiting adjuncts (*upādhi*); see Deutsch (1969) and Potter (1981) for scholarly introductions. The short treatise *Dṛg-Dṛśya-Viveka*, traditionally attributed to Śaṅkara (an attribution not secure on historical grounds), analyses the distinction between seer and seen (Nikhilananda, 1931). *Samādhi* is treated in the *Yoga Sūtras* of Patañjali (Bryant, 2009). We take these texts as a conceptual source, not as empirical evidence.

3. Formal Framework

3.1 Primitive objects

Definition 1 (Reality field)**.** The *reality field* is a function $f : \mathbb{R} \to \mathbb{R}$ (or, more generally, $f : \mathbb{R}^n \to \mathbb{R}^m$), assumed locally integrable and continuously differentiable where required. It is *undivided*: it carries no internal boundaries, no distinguished centre and no internal friction. In Advaita terminology it corresponds to *Akhaṇḍam* (the undivided) and, as seer, to *Drig*.

Definition 2 (Observer-filter)**.** An *observer-filter* centred at $x_i \in \mathbb{R}$ is a measurable, bounded function $w_i(x) = w(x - x_i)$, $0 \le w_i \le 1$, called the *windowing function*. It encodes the physical sense organs, cognitive structures, conditioning, beliefs and language of the observer. It corresponds to *upādhi*, the limiting adjunct.

Definition 3 (Observed fragment)**.** The *fragment* observed through w_i is

$$S_i(x) = f(x) \cdot w(x - x_i). \tag{1}$$

It corresponds to *Dṛśyam*, the seen.

Definition 4 (Reconstruction and cognitive delta)**.** Given two fragments S_i, S_j with overlap $\Omega_{ij} = \operatorname{supp} w_i \cap \operatorname{supp} w_j$, the mind forms a reconstruction $\hat{f} = T_{ij}(S_i, S_j)$ via a transformation T_{ij}. The *cognitive delta* of the pair is the boundary mismatch

$$\Delta_{ij} = \|\nabla S_i - T(\nabla S_j)\|_{L^2(\Omega_{ij})}\,. \tag{2}$$

3.2 Axioms

Axiom 1 (Undivided source [P])**.** There exists an undivided field f whose internal friction is identically zero.

Axiom 2 (Observation requires a filter [P])**.** Any observation of f is mediated by some observer-filter $w_i \not\equiv 1$ centred at a bounded location x_i, so that what is observed is S_i and never f itself.

Axiom 3 (Position-dependence [P])**.** Distinct observers occupy distinct centres and generally possess distinct filters, $w_i \neq w_j$.

Axiom 4 (Cost of boundaries [P])**.** Maintaining a boundary or a forced match between incompatible fragments requires a continuous supply of energy (metabolic, psychological or egoic), and this cost increases with the mismatch.

3.3 Operational causal chain

The theory describes the following chain, depicted in Figure 1.

1. **Continuous reality** $f(x)$: the undivided, self-consistent field.
2. **Observation as filter** $w(x - x_i)$: the bounded point of view introduced at x_i.
3. **Filtered fragment** $S_i = f(x) \cdot w(x - x_i)$.
4. **Mismatched interlocks:** the attempt to stitch incompatible fragments S_i, S_j.
5. **Cognitive delta** Δ: the mathematical error signal of boundary mismatch, experienced as friction, anxiety and dogma.

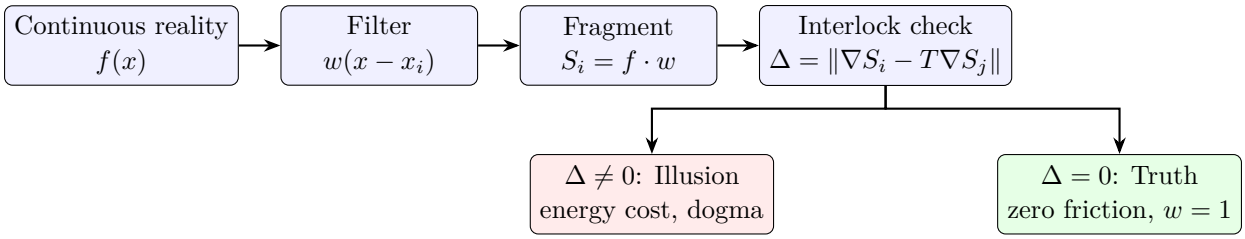


Figure 1: The decoding pipeline of Filter Theory: from undivided source to bifurcation of truth ($\Delta = 0$) versus illusion ($\Delta \neq 0$).

4. Primary Illusion: Observation as a Filter Function

4.1 The three artificial constructs

Classical epistemology often treats perception as a passive window onto an external reality. Filter Theory instead treats perception as an *active manufacturing of separation*. To observe any object or state, three conditions must be instantiated simultaneously:

1. **A bounded point** x_i: the assertion of a spatial, temporal or egoic location ("I am located here").
2. **A primary separation:** the creation of duality ("the observed is over there, distinct from me").
3. **A filtering function** $w(x)$: an admission mechanism that accepts only specific frequencies or parameters.

The moment an observer asserts "I see", a boundary is imposed where there was none, duality where there was unity, and selection where there was completeness. The filter $w(x)$ is not a neutral instrument that allows reality to be seen; in this model it is precisely what *conceals* reality while appearing to display it.

4.2 Mathematical formulation of perception

The observed phenomenon is never raw reality $f(x)$ but a windowed fragment S_i given by (1), whose constituents are summarised in Table 1.

Symbol	Meaning in the model	Advaita correspondent
$f(x)$	Continuous underlying reality: undivided, self-consistent, infinite, free of internal boundaries	*Akhaṇḍam*; *Drig*; *Brahman*
$w(x - x_i)$	Observer's windowing function: eye, sense organs, cognitive structures, conditioning, beliefs, language, centred at x_i	*Upādhi*
x_i	Centre of observation: the spatial, mental or egoic point of view	*Ahaṃkāra* (ego-sense) as the causal-level centre
S_i	Observed fragment: always a localised product of reality and filter	*Dṛśyam*
Δ	Boundary mismatch between fragments from different filters	*Avidyā*-generated friction

Table 1: Symbols of Filter Theory and their Advaita correspondents.

4.3 Parallels in Advaita and quantum mechanics [A]

Two parallels motivate the model.

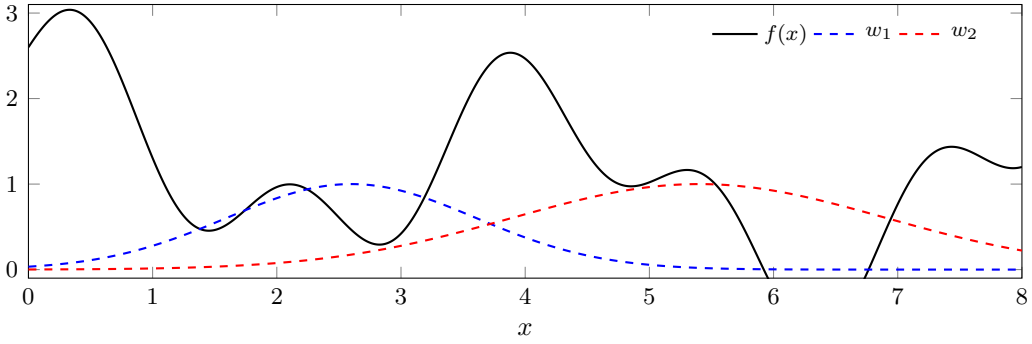
- **Advaita Vedānta (*upādhi*).** Consciousness (*cit*) adopts a limiting adjunct, the mind, in order to perceive. In seeing through an *upādhi*, the undivided *Brahman* appears divided into subject and object.
- **Quantum measurement.** A quantum system remains in a coherent superposition until measured; measurement requires a cut between observer and system, and the cut influences which outcome is observed (von Neumann, 1932; Zurek, 2003).

These are structural analogies: in both, the *partition into observer and observed is a precondition of an observation and shapes its result.* We do not claim that quantum measurement entails Advaita, or conversely.

5. Topology of Separation and Fragmented Sampling

Prior to observation, reality exists as a continuous field $f(x)$ without edges, centre points or internal friction ($\Delta = 0$). Once observation occurs, reality is fragmented into a collection of

(a) Reality $f(x)$ and two observer windows w_1, w_2



(b) Fragments $S_1 = f \cdot w_1$ and $S_2 = f \cdot w_2$

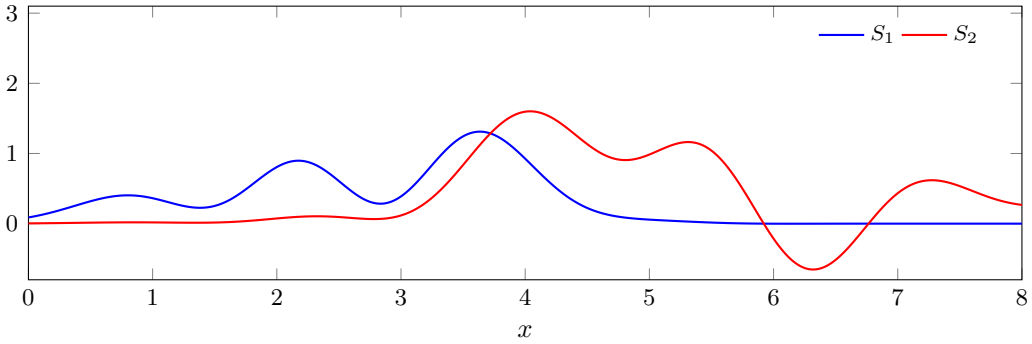


Figure 2: Illustration of Eq. (1). A single field $f(x) = \cos 1.5x + 0.6 \sin 3.7x + 0.4 \cos 0.5x + 1.2$ is observed through two Gaussian filters, w_1 (centre 2.6, width 1.0) and w_2 (centre 5.4, width 1.5). Each fragment is an authentic part of f yet bears the imprint of its own window; neither equals f.

localised samples

$$\text{Observed Domain} = \{S_1, S_2, S_3, \ldots, S_n\}. \tag{3}$$

Each fragment S_i carries the indelible imprint of the specific filter $w(x - x_i)$ that generated it.

5.1 The cup and the ocean

Dip a cup into the ocean. The water in the cup is authentic ocean water with exactly the sea's chemical composition. Yet the rigid geometry of the cup (the filter) induces the illusion that the ocean itself is cup-shaped, bounded and isolated. If several observers dip differently shaped cups, each holds authentic water, but the container shapes differ. Attempting to reassemble the ocean by stitching the *shapes* of the cups together is impossible, because their boundary geometries do not match. The content is genuine; the boundary is the artefact.

5.2 Three structural properties of filters

1. **Band-limiting.** A filter admits only a narrow band of the available information. The human eye responds only to electromagnetic wavelengths of roughly 400–700 nm, and the mind admits only concepts that align with its established cognitive structures.

2. **Position-dependency.** A filter is anchored at x_i. Personal history, physical body, trauma, culture and language fix the coordinates of x_i. Two observers at x_i and x_j sample the same $f(x)$ yet produce fragments $S_i \neq S_j$; each, on the strength of an authentic sample, may claim absolute truth.

3. **Energy consumption.** A boundary does not maintain itself. Continuous metabolic, psychological and egoic energy is expended in asserting "I am this bounded self, and not that" (Axiom 4).

5.3 Inequality of the sum and the whole

The sum of localised, filtered observations does not in general equal the undivided whole. We make the statement exact.

Proposition 1 (Recoverability of the whole from fragments [M])**.** *Let* $w_1, \dots, w_n \in L^\infty(\mathbb{R})$ *and* $W = \sum_{i=1}^n w_i$*. Then*

$$\sum_{i=1}^{n} \int_{\mathbb{R}} S_i(x)\,\mathrm{d}x = \int_{\mathbb{R}} f(x)\,\mathrm{d}x \quad \textit{for every } f \in L^1(\mathbb{R})$$

if and only if $W(x) = 1$ *for almost every* x *(that is,* $\{w_i\}$ *is a partition of unity).*

Proof. We have $\sum_i \int S_i = \int fW$, so the identity is equivalent to $\int f(1-W) = 0$ for all $f \in L^1$. If $W = 1$ a.e. this holds trivially. Conversely, suppose $1-W \neq 0$ on a set of positive measure; then there is a bounded set A of positive measure on which $1-W$ has a fixed sign and $|1-W| \geq \varepsilon > 0$. Taking $f = \mathbf{1}_A$ gives $\int f(1-W) \neq 0$, a contradiction. □

Corollary 1. *If every* w_i *has bounded support (a finite observer with a bounded point of view) and* $n < \infty$*, then* W *vanishes outside a bounded set, so* $W \neq 1$ *on a set of infinite measure, and there exist fields* f *whose integral cannot be recovered from the fragments.*

Remark 1 (Relation to the transparent filter). The special case $n = 1$, $w \equiv 1$ is the *transparent* filter: no window, no isolated observer, no separation. The original statement of the theory, "the sum equals the integral if and only if $w = 1$ everywhere", is the one-observer instance of Proposition 1. The general condition is the weaker partition-of-unity requirement, which a family of many cooperating, jointly exhaustive observers could in principle meet; Corollary 1 shows that this is unattainable for finitely many bounded observers. This refinement does not weaken the thesis, because actual observers are bounded, finite, and in addition carry *observer-specific* (non-identical) windows, which is what produces the mismatch of Section 6.

Definition 5 (Level 1 illusion)**.** *Level 1 illusion* is the unexamined belief that $S_i = f(x)$: the mistaking of a filtered fragment for the whole.

Remark 2 (Beyond multiplicative windows). The multiplicative model (1) is the simplest member of a larger class $S_i = \mathcal{W}_i[f]$ in which $\mathcal{W}_i$ is a (possibly nonlinear, noisy or history-dependent) operator representing the entire sensory–cognitive pipeline. All qualitative conclusions below concern properties of such operators (boundedness, non-identity across observers); the explicit multiplicative form is used because it permits closed-form results.

6. Secondary Illusion: Mismatched Interlocks and Cognitive Delta

If Level 1 illusion is the root of perception, Level 2 illusion is the tangled tree that grows from it. Once consciousness possesses multiple fragments S_i, S_j, the mind attempts to reconstruct the continuous field through a transformation T_{ij}:

$$\hat{f}(x) = T_{ij}(S_i, S_j). \tag{4}$$

6.1 Truth condition versus illusion condition

For two observations to be joined without internal tension, their edge gradients must agree across the shared boundary:

$$\text{Truth condition:} \quad \nabla S_i = T(\nabla S_j) \implies \Delta = 0 \quad \text{(zero friction).} \tag{5}$$

Since each fragment bears the imprint of a distinct filter, $w(x - x_i) \neq w(x - x_j)$, the boundaries are in general incompatible:

$$\text{Illusion condition:} \quad \|\nabla S_i - T(\nabla S_j)\| = \Delta \neq 0. \tag{6}$$

The following proposition makes the link between filter difference and mismatch precise for the simplest transformation, the identity.

Proposition 2 (Mismatch characterisation [M])**.** *Let $\Omega = \Omega_{ij}$ be a bounded, connected open interval, let $w_i, w_j \in C^1(\Omega)$, and let $T = \mathrm{id}$. Then:*

(a) $\nabla S_i - \nabla S_j = \nabla\big(f\,(w_i - w_j)\big)$ on Ω, hence $\Delta_{ij} = 0$ for a given $f \in C^1(\Omega)$ if and only if $f\,(w_i - w_j)$ is constant on Ω.

(b) $\Delta_{ij} = 0$ for every *$f \in C^1(\Omega)$ if and only if $w_i = w_j$ on Ω.*

Proof. (a) By linearity, $\nabla S_i - \nabla S_j = \nabla(S_i - S_j) = \nabla\big(f(w_i - w_j)\big)$. Its $L^2(\Omega)$ norm vanishes iff the derivative vanishes a.e., i.e. iff the continuous function $f(w_i - w_j)$ is constant on the interval Ω.

(b) If $w_i = w_j$ on Ω the difference is identically 0. Conversely, put $g = w_i - w_j$. Taking $f \equiv 1$ in (a) shows g is a constant c; taking $f(x) = x$ gives $f\,g = cx$, which is constant on Ω only if $c = 0$. □

Remark 3. Proposition 2 formalises the statement that Δ *is not an error in reality*: f is unbroken and enters only through the product $f \cdot g$. Δ is mathematical evidence that the two active filters do not agree. If $w_i \equiv w_j \equiv 1$ (the transparent filter), then $\Delta = 0$ for every f, the limiting case $w \to 1$ of Section 8.

A worked example. For the field and the two Gaussian filters of Figure 2, the overlap region $[3, 5]$ yields (numerically, with $T = \mathrm{id}$) $\Delta_{12} \approx 1.78$. Figure 3 displays the gradient of each fragment on the overlap and the residual that constitutes Δ. The mismatch is nonzero even though both fragments are exact samples of the same f.

6.2 The nature and phenomenology of delta

In human psychological experience, Δ is hypothesised to manifest as:

- cognitive dissonance and inner contradiction (Festinger, 1957);
- the compulsive need to rationalise, justify and tell self-serving stories;
- psychological strain, chronic anxiety and hypervigilance;
- the demand for rigid external beliefs, dogmas and ideologies to hold conflicting worldviews together.

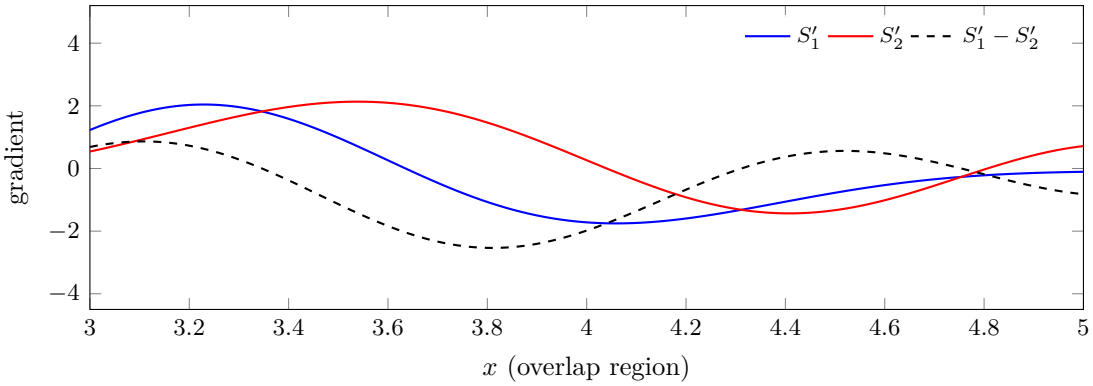


Figure 3: Boundary mismatch on the overlap $[3, 5]$ of the two fragments of Figure 2. The dashed curve is the pointwise residual whose L^2 norm is $\Delta_{12} \approx 1.78$.

This correspondence is a hypothesis [P] to be tested (Section 11); the mathematical object Δ is defined independently of it.

6.3 Mechanics of forced interlocking

1. **Mechanical deformation.** The observer alters or suppresses the edges of S_i to fit S_j. Psychologically this corresponds to selective denial, self-deception and compartmentalisation ("I will ignore this contradictory evidence so my worldview remains intact").
2. **Continuous energy supply.** Incompatible edges repel, so holding them together requires a continuous supply of energy. We formalise this as follows.

Postulate 1 (Cost of forced interlock [P])**.** The power required to hold a forced match between fragments is an increasing function of the mismatch; in the simplest model $E_{ij} = \kappa\, \Delta_{ij}^2$ with a coupling constant $\kappa > 0$ characteristic of the observer. Total cost is $E = \sum_{i<j} E_{ij}$.

Belief is the force applied; dogma is the battery. The greater the mismatch, the greater the required energy: fanaticism is expensive; truth, for which $\Delta = 0$ and hence $E = 0$, is cost-free.

3. **Cascade errors.** A forced interlock becomes the foundation for future observations: each subsequent sample S_k must be deformed to match an already distorted reconstruction $\hat{f}$. The next proposition quantifies this compounding.

Proposition 3 (Error cascade [M])**.** *Let the exact sequential reconstruction be $f_k^\star = T_k(f_{k-1}^\star, S_k)$ and the deformed one be $\hat{f}_k = T_k(\hat{f}_{k-1}, S_k) + \eta_k$, with common start $\hat{f}_0 = f_0^\star$, where η_k is the fresh deformation introduced by forcing the k-th fragment to fit, $\|\eta_k\| \le \varepsilon_k$. Suppose each T_k is L-Lipschitz in its first argument and put $e_k = \left\|\hat{f}_k - f_k^\star\right\|$. Then*

$$e_n \;\le\; \sum_{k=1}^{n} L^{\,n-k}\, \varepsilon_k .$$

In particular, if $L > 1$ and $\varepsilon_k \ge \varepsilon > 0$, the bound grows geometrically with n, whereas for $L < 1$ and $\varepsilon_k \le \varepsilon$ it stays below $\varepsilon/(1 - L)$.

Proof. $e_1 \le \|\eta_1\| \le \varepsilon_1$. For $k \ge 2$, $e_k \le \left\|T_k(\hat{f}_{k-1}, S_k) - T_k(f_{k-1}^\star, S_k)\right\| + \|\eta_k\| \le L e_{k-1} + \varepsilon_k$, and induction gives the stated sum. For $L < 1$ the sum is dominated by $\varepsilon \sum_{m \ge 0} L^m = \varepsilon/(1-L)$. □

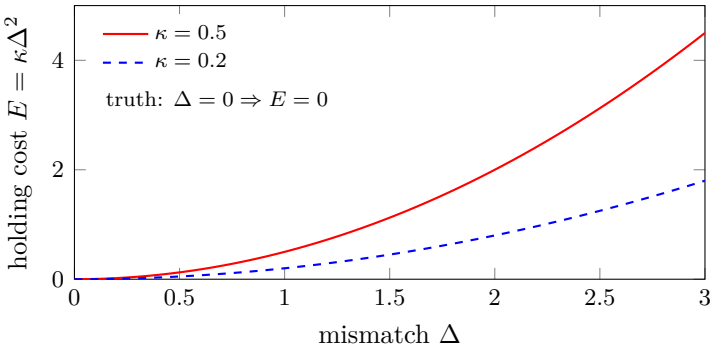


Figure 4: Postulated cost of holding mismatched fragments together (Postulate 1). Observers with greater coupling κ pay more for the same mismatch.

6.4 Truth versus illusion: summary

Dimension	Truth ($w = 1$)	Illusion ($\Delta \neq 0$)
Filter state	Transparent / unbounded ($w = 1$)	Opaque / windowed ($w_i \neq w_j$)
Boundary fit	Seamless: $\nabla S_i = T(\nabla S_j)$	Mismatched: $\|\nabla S_i - T(\nabla S_j)\| = \Delta$
Energy required	Zero energy; effortless	High energy; dogma; force
Psychological state	Relaxation; samādhi; peace	Cognitive dissonance; anxiety
Structural effect	Undivided whole $f(x)$	Fragmented cascades $\hat{f}(x)$

Table 2: Comparison of the truth condition and the illusion condition.

7. Signal-Processing and Information-Theoretic Analogies

The limitations of perception can be modelled with standard results from signal processing and information theory. Throughout this section the statements about mathematics are theorems [M] and the identification of cognition with the mathematical objects is an analogy [A].

7.1 Nyquist–Shannon sampling

A signal g whose Fourier transform is supported in $[-B, B]$ is determined by its samples at rate $f_s > 2B$ (the Nyquist condition); exact reconstruction uses the Whittaker–Shannon interpolation formula and in principle requires infinitely many samples on an unbounded domain (Nyquist, 1928; Shannon, 1949). Real observers sample on a bounded domain at a finite rate, and a physical field need not be band-limited at all. Hence perfect reconstruction of f from an observer's sensory and mental samples is not guaranteed and, for unbounded-bandwidth f, impossible by design.

The degradation pipeline has three stages: the continuous source $f(x)$ (potentially infinite bandwidth) $\to$ a band-limited, spatially bounded filter $w(x) \to$ an undersampled fragment S_i carrying reconstruction error Δ.

7.2 Windowing, uncertainty and the capacity of a filter

Multiplication in space corresponds to convolution in frequency: $\widehat{S_i} = \hat{f} * \widehat{w_i}$. This has two consequences.

Spectral leakage. A spatially bounded window cannot have a compactly supported spectrum: a nonzero function cannot be both time-limited and band-limited (a consequence of the Paley–Wiener theorem). "Strict band-limiting" and "strict spatial boundedness" of a cognitive filter are therefore not simultaneously attainable; we should say that a filter is *effectively* concentrated in both domains (Slepian and Pollak, 1961; Harris, 1978).

Finite capacity. For signals effectively concentrated on an interval of length T and a band of width $2B$, the number of degrees of freedom is approximately $2BT + 1$ (Slepian and Pollak, 1961). A bounded observer therefore has a *finite-dimensional* view of an infinite-dimensional field. We denote this capacity $N_i \approx 2B_iT_i$, and note that two observers with different (B_i, T_i) and centres have different, only partly overlapping, subspaces, which is the signal-theoretic version of position-dependency.

7.3 The mind as a lossy codec

The cognitive apparatus functions as a lossy compression codec: it compresses the high-dimensional field into low-bandwidth fragments suited to survival and biological efficiency. In rate–distortion theory, for a memoryless Gaussian source of variance σ^2 under squared-error distortion, the minimum achievable distortion at rate R bits per sample is

$$D(R) = \sigma^2\, 2^{-2R}, \tag{7}$$

which is strictly positive for every finite R and tends to zero only as $R \to \infty$ (Shannon, 1959; Cover and Thomas, 2006). Finite channel capacity thus guarantees nonzero distortion, a quantitative counterpart to the claim that finite bounded observers cannot achieve $w = 1$.

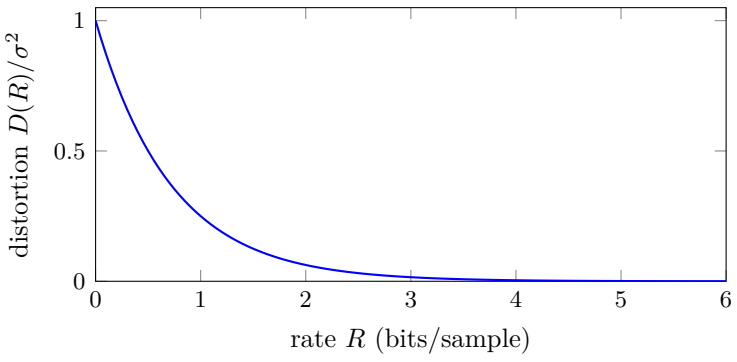


Figure 5: Gaussian rate–distortion function (7). Distortion decays exponentially with rate but never reaches zero at finite rate.

Efficiency is vital for biological survival, but efficiency is not truth. In this light, wisdom is not the construction of a better codec but the *knowledge that one is using a codec.*

Domain	Established result	Role in Filter Theory
Sampling theory	$f_s > 2B$ suffices for band-limited signals; infinite samples for exact reconstruction	Finite sampling of a rich field yields an irreducible error Δ
Uncertainty / windows	No nonzero function is both time- and band-limited; capacity $\approx 2BT$	A bounded observer sees a finite-dimensional subspace
Rate–distortion	$D(R) > 0$ for finite R	Compression is lossy; efficiency $\neq$ truth
Quantum measurement	Outcome depends on system–apparatus cut	Observer–observed partition shapes what is seen
Cognitive dissonance	Inconsistency among cognitions produces tension	$\Delta \neq 0$ at fragment boundaries

Table 3: Cross-disciplinary correspondences. Each row is an analogy [A], not an identity.

7.4 Summary of analogies

8. Deconstruction: The Four-Step Protocol to Samādhi

Conventional self-help, therapeutic models and philosophical systems attempt to repair Level 2 illusions by improving the transformation T: better beliefs, refined coping mechanisms, reshaped narratives. Filter Theory addresses Level 1 directly, asserting that the observer-filter itself is optional in its limiting state.

8.1 The four steps

Step 1. Identify $w(x)$. Locate and name the active filter. Shift the inquiry from "What am I seeing?" to "*Through what filter* am I seeing?" The filter may be fear, conditioning, grief, language or a specific mental tendency. A filter that is seen loses its power to blind.

Step 2. Cease force (*neti neti*). Stop supplying energy to hold mismatched edges together. Apply the classical Vedāntic negation *neti neti* ("not this, not this"). Allow fragment S_i to exist purely as a filtered sample without forcing it to represent total reality $f(x)$. In model terms: reduce the term $E = \kappa\Delta^2$ by ceasing to enforce the match.

Step 3. Decouple S_i. Allow fragments S_i and S_j to separate naturally and rest in the unstitched state. This initially feels like vertigo, groundlessness or confusion. In the model this is an *accurate* perception: the "ground" was an artefact of forced stitching, and its dissolution is felt as loss of ground.

Step 4. Re-align $w \to 1$ by non-observation. As effort and force are withdrawn, the filter relaxes. $f(x)$ need not be constructed or reconstructed; it is disclosed when the blocking mechanism ceases.

8.2 Definition of samādhi in the model

Definition 6 (Samādhi, model sense [P])**.** *Samādhi* is the limiting state of complete filter relaxation:

$$w \equiv 1, \qquad x_i \text{ dissolved}, \qquad S_i = f(x), \qquad \Delta \equiv 0, \qquad E = 0,$$

a state characterised by zero friction, zero effort and total transparency.

8.3 The equation of liberation

Liberation is achieved not by superior edge-stitching but by recognising the filter as a filter rather than as reality:

$$w(x) \to 1 \implies S_i \to f(x), \qquad \Delta \to 0. \tag{8}$$

Proposition 4 (Limit of liberation [M])**.** *Let $f \in C^1(\mathbb{R})$ and let $w^{(\lambda)} \in C^1(\mathbb{R})$, $0 \le w^{(\lambda)} \le 1$, be a family of windows such that, as $\lambda \to \infty$, $w^{(\lambda)} \to 1$ and $\nabla w^{(\lambda)} \to 0$ locally uniformly (for example $w^{(\lambda)}(x) = e^{-(x-x_i)^2/2\lambda^2}$). Then $S^{(\lambda)} = f\, w^{(\lambda)} \to f$ and $\nabla S^{(\lambda)} \to \nabla f$ locally uniformly. Consequently, for two such families centred at distinct x_i, x_j, the mismatch Δ_{ij} over any fixed bounded overlap tends to 0.*

Proof. On a compact set K, $\sup_K |f(w^{(\lambda)} - 1)| \le \sup_K |f| \cdot \sup_K |w^{(\lambda)} - 1| \to 0$. Moreover $\nabla S^{(\lambda)} - \nabla f = (w^{(\lambda)} - 1)\nabla f + f \nabla w^{(\lambda)} \to 0$ uniformly on K. For two families, $\nabla S_i^{(\lambda)} - \nabla S_j^{(\lambda)} \to \nabla f - \nabla f = 0$ uniformly on the compact overlap, so its L^2 norm tends to 0. □

Proposition 4 confirms that the limit (8) is mathematically consistent for the Gaussian family: as the window broadens to the whole domain, the transformation T becomes unnecessary and the truth condition (5) holds automatically.

8.4 Decoding pipeline and edge geometry

- **Truth (complementary edges, $w = 1$).** The edge of S_1 fits the edge of S_2 seamlessly; no force is required and energy expenditure is zero.
- **Illusion (non-matching edges, $w_1 \neq w_2$).** The edge of S_1 diverges from the edge of S_2; the resulting gap is Δ, representing boundary failure and psychological resistance. To bridge it the ego must continuously supply belief-energy.

9. Classical Foundations: *Dṛg-Dṛśya-Viveka* as Filter Theory

The mathematical formulation of Filter Theory aligns structurally with the classical treatise *Dṛg-Dṛśya-Viveka* ("Discrimination between the Seer and the Seen"), traditionally attributed to Śaṅkara (Nikhilananda, 1931). The correspondence below is offered as an interpretive mapping, not as a claim about the authors' intentions.

9.1 Core principle and equation mapping

The opening verse states, in substance, that whatever is seen is *dṛśya* by virtue of being seen, and that what is seen cannot be the seer (*dṛk*). The mapping is as follows.

- ***Dṛśyam*(the seen):** the bounded, changing, filtered fragment $S_i = f(x) \cdot w(x - x_i)$.
- ***Drig*(the seer):** the unbounded, unchanging, filter-less continuous reality $f(x)$.
- ***Viveka* (discrimination):** the clear discernment that separates *Drig* from *Dṛśyam*.
- ***Upādhi* (limiting adjunct):** the filter $w(x)$ that makes the infinite *Drig* appear as a finite *Dṛśyam*.

9.2 Three hierarchical levels of *Dṛśyam*

1. ***Sthūla dṛśyam* (gross seen):** the physical body, external objects and sensory environments (S_i). The ego falsely claims, "I am the body".
2. ***Sūkṣma dṛśyam* (subtle seen):** thoughts, emotions, concepts and the filter $w(x)$ itself. Crucially, because the mind's filter can itself be observed, *the filter is also dṛśyam (S_j), not the seer.*
3. ***Kāraṇa dṛśyam* (causal seen):** the core habit of identification, the primal act of establishing x_i as a centre. The subtle assertion "I am located at x_i" is the deepest filter.

9.3 The negative equation of viveka

To isolate the true seer, the practitioner applies a logical elimination. Let $\mathcal{O}$ denote the set of observable items.

$$X \in \mathcal{O} \implies X \neq \textit{Drig}. \tag{9}$$

Applied successively: *body* is observable ⇒ body ≠ *Drig*; *thought* is observable ⇒ thought ≠ *Drig*; *filter* $w(x)$ is observable ⇒ $w(x) \neq$ *Drig*; *centre* x_i is observable ⇒ $x_i \neq$ *Drig*; *ego* ("I am the seer") is observable ⇒ ego ≠ *Drig*.

What remains when all observable objects, filters and centres are negated is that which cannot be negated: the self-luminous (*svayam-prakāśa*) unobserved knower, *Drig*, identified in the model with $f(x)$.

Remark 4 (Logical status). The elimination (9) is valid as an inference only if the premise "the observable is not the observer" is granted; it is a premise of the Advaita tradition and is related to, but not identical with, the epistemological regress arguments familiar from Western philosophy of mind. The residue of the elimination is a *postulate* [P] in the framework, and the identification of *Drig* with the mathematical field f is a modelling choice.

Level	Sanskrit term	Filter-Theory object
Gross	*Sthūla dṛśyam*	Fragments S_i of the physical environment
Subtle	*Sūkṣma dṛśyam*	Thoughts, emotions and the filter $w(x)$ itself
Causal	*Kāraṇa dṛśyam*	The centre x_i (act of identification)
Residue	*Drig*	The field $f(x)$ as unobserved knower

Table 4: Hierarchical levels of the seen, and the residue of elimination.

10. Practical *Sādhanā* and the Dynamics of Grace

10.1 The daily four-question inquiry

Whenever friction, anxiety, mental strain or the urge to rationalise arises ($\Delta \neq 0$), the practitioner poses four sequential questions.

Q1. *What is being seen right now?* (*Kiṃ dṛśyam?*) Recognise the anxiety or thought as a fragment S_i.

Q2. *Through what filter is it being seen?* (*Kena upādhinā dṛśyam?*) Recognise the filter $w(x-x_i)$ as *Dṛśyam*.

Q3. *Where is the centre that claims to see?* (*Kutra x_i?*) Recognise the centre x_i itself as a sensation or appearance, hence *Dṛśyam*.

Q4. *What knows all three?* (*Kiṃ dṛk?*) Rest as the unobserved knower: be *Drig*.

10.2 Intended results

- The fragment S_i remains but is no longer mistaken for total reality $f(x)$.
- The filter $w(x)$ is recognised as a functional tool, not an identity.
- The centre x_i dissolves from an absolute identity into a transient appearance.
- Δ collapses as energy is withdrawn from holding mismatched edges (Postulate 1).

10.3 Divine grace (*anugraha*) and vertigo as revelation

In classical spirituality, grace is often understood as an external intervention. Within Filter Theory it is reinterpreted as the *unearned availability of* $w = 1$: continuous reality $f(x)$ disclosing itself through the filter despite the active presence of $w(x)$. Grace requires no intermediary filter and no effort, being free because *Drig* is eternally present and inherently unconstrained. When the centre x_i is observed as *Dṛśyam*, the perceived ground of identity vanishes. The ensuing vertigo is, in this reading, not a failure but a disclosure of the "groundless ground" of *Drig*.

Remark 5 (Safety note). Because the protocol may provoke strong feelings of groundlessness or derealisation, practitioners with a history of dissociation, psychosis or severe anxiety should engage only with qualified clinical or contemplative guidance. This framework is a philosophical model and is not a medical or therapeutic treatment.

11. Empirical Programme and Falsifiability

A theory that is purely philosophical is judged by coherence and explanatory scope. The mathematical core, however, supports concrete, falsifiable predictions on the psychological side. We propose the following programme.

H1. Mismatch and dissonance. Operationalise Δ in perceptual tasks, using two observers (or one observer under two filter conditions, e.g. differing spatial frequency bands) viewing

overlapping parts of the same stimulus, and define Δ as the boundary-gradient disagreement of their reports. *Prediction:* self-reported dissonance and physiological arousal (skin conductance, heart-rate variability) increase monotonically with Δ.

H2. Cost scaling. If Postulate 1 is correct, effortful reconciliation (measured by reaction time, pupil dilation, or blood-oxygen-level-dependent activity in conflict-monitoring regions) should scale approximately quadratically with Δ. *Falsifier:* a linear or saturating relationship in a well-powered study.

H3. Cascade. In sequential-integration tasks, error in the final integrated estimate should grow with the number of forced reconciliations according to Proposition 3; compare against Bayesian sequential integration, for which no geometric growth is expected.

H4. Filter identification. Training that directs attention to the active filter ("through what am I seeing?") should reduce measured reconciliation cost relative to a control that trains better stitching (belief revision). This tests the claim that Level 1 intervention outperforms Level 2 intervention for dissonance reduction.

H5. Contemplative states. Experienced practitioners reporting states close to the defined samādhi should show reduced indices of effortful self-referential processing (for example, reduced default-mode-network activity), a hypothesis consistent with existing contemplative-neuroscience work but requiring direct testing against this model's specific predictions.

We emphasise that H1–H5 are *proposals*; no empirical result is claimed in this paper.

12. Discussion and Limitations

12.1 Interpretive scope

Filter Theory can be read at three increasingly strong levels:

(i) **Epistemic (weak):** all human observation is partial and perspective-bound, and disagreements arise partly from non-matching perspectives. This is widely accepted.

(ii) **Psychological (medium):** reconciling incompatible perspectives is effortful, and the effort has costs that can be reduced by recognising one's filters. This is empirically testable (Section 11).

(iii) **Metaphysical (strong):** an undivided field exists and is disclosed when the filter is fully relaxed. This is a philosophical postulate of Advaita and cannot be established by the mathematics of this paper.

The mathematical results of Sections 4–8 hold at all three levels, as they concern only the formal model.

12.2 Limitations

1. **Self-reference.** The theory is itself a fragment S produced by some filter. By its own criterion it cannot claim to be f. The framework accepts this: it positions itself as a *pointer* that is to be dissolved on use, not a final description. Critics may reasonably regard this as either a strength (consistency) or a weakness (unfalsifiability at the strong level).

2. **Modelling simplifications.** The multiplicative window, identity transformation T and the L^2 norm in Proposition 2 are simplifications. Real perception involves nonlinear, noisy, prediction-laden processing (Friston 2010); the generalisation $S_i = \mathcal{W}_i[f]$ preserves the qualitative argument but not the closed-form results.

3. **Analogies are not derivations.** Correspondences with quantum mechanics, sampling theory and Advaita are structural and heuristic. The quantum measurement problem has competing interpretations, and not all of them assign an essential role to an observer.

4. **The energy postulate.** Postulate 1 is a modelling assumption; the quadratic form is chosen for tractability.

5. **Samādhi as a limit.** The state $w \equiv 1$ is a mathematical limit; whether it corresponds to any attainable experiential state, and whether the reports of practitioners are veridical, are matters on which this paper takes no empirical position.

6. **Textual scholarship.** The attribution of *Dṛg-Dṛśya-Viveka* to Śaṅkara is traditional; scholarship is divided on its authorship and date, and our reading is one of several possible.

12.3 Relation to prior work

Relative to reduction-valve and interface theories, the contribution of Filter Theory is a precise geometric account of mismatch between observers, a quantitative cost model, and a protocol. Relative to cognitive dissonance theory, it provides a formal object Δ with an explicit dependence on the difference of filters. Relative to Advaita scholarship, it provides a modern formal vocabulary for the classical seer–seen analysis, with its metaphysical content flagged explicitly as postulate.

13. Conclusion

We have presented Filter Theory, in which perception is modelled as the windowing $S_i = f(x) \cdot w(x - x_i)$ of an undivided field. Primary illusion is the mistaking of a fragment for the whole; secondary illusion is the mismatch Δ that results when fragments from different filters are forcibly joined. We proved that finitely many bounded observers cannot recover the whole from their fragments (Proposition 1), that mismatch vanishes for all fields iff the filters agree on their overlap (Proposition 2), that forced sequential reconciliation can compound error (Proposition 3), and that the limit $w \to 1$ is mathematically consistent (Proposition 4). We related these to sampling, uncertainty, rate–distortion and measurement, grounded them in *Dṛg-Dṛśya-Viveka*, and proposed an empirical programme.

The practical thrust may be summarised in a sentence: *reality does not need to be constructed; it needs to be allowed*. Illusions are costly to maintain, since they must be paid for in energy, whereas truth, requiring no energy, is free; and observation, in this model, is the invoice.

Quantity	Formula	Meaning
Realised reality	$f(x)$	Continuous, undivided, infinite field (*Brahman* / *Drig*)
Observer filter	$w(x - x_i)$	Windowing function / *upādhi* centred at x_i
Observed fragment	$S_i = f(x)\, w(x - x_i)$	Filtered observation / *Dṛśyam*
Reconstruction attempt	$\hat{f}(x) = T_{ij}(S_i, S_j)$	The mind's attempt to stitch fragments via T
Boundary friction	$\Delta = \|\nabla S_i - T(\nabla S_j)\|$	Evidence of filter mismatch / cognitive dissonance
Holding cost	$E = \kappa \Delta^2$	Postulated energy cost (Postulate 1)
Truth condition	$\Delta = 0 \iff w_i = w_j$ on Ω_{ij}	Zero energy, seamless fit (Prop. 2)
Recoverability	$\sum_i w_i = 1$ a.e.	Partition of unity (Prop. 1)
Equation of viveka	$X \in \mathcal{O} \Rightarrow X \neq Drig$	Elimination of body, mind, filter, centre
Equation of liberation	$w \to 1 \Rightarrow S_i \to f,\ \Delta \to 0$	Dissolution of centre x_i into undivided *Drig*

Table 5: Master summary of the principal formulas.

A. Master Summary of Mathematical Formulations

B. English Rendering of the Malayalam Synthesis

The original draft contained a native Malayalam synthesis (Section 9, *"Dṛk-Dṛśya Viveka: Observation Itself Is Māyā"*). For the purposes of an English-language publication, its content is rendered below; the Malayalam text is preserved in the source draft.

B.1 Main thesis. Observation itself is the root cause of *māyā*. Three things are indispensable for observing an object: (1) a bounded viewpoint, the awareness "I am here (x_i)"; (2) separation, the duality "the object seen is over there"; and (3) a filter $w(x)$, the limitation "only what passes through this filter shall I see" (eye, mind, language, belief). With $S_i = f(x)\, w(x - x_i)$: $f(x)$ is the continuous fundamental reality (undivided and indivisible); $w(x)$ is the observer's filter (*upādhi*: eye, mind, experience); S_i is the fragment we see (*dṛśyam*). We never see $f(x)$; we see only S_i. At the very moment of saying "I see" we create a boundary (where none existed), duality (where there was unity) and selection (where there was completeness).

B.2 Resolution (release from the filter).

1. Identify the filter: ask not "What do I see?" but "*Through what* do I see?"

2. Stop applying force (*neti neti*): do not spend energy holding the false edges together.
3. Separate: allow the fragments to separate by themselves. In this state, vertigo (groundlessness) is felt; that is the correct seeing.
4. When the filter dissolves ($w \to 1$): the picture need not be constructed; it reveals itself. This is samādhi.

B.3 Conclusion. Reality need not be constructed; it must be allowed. Illusions must be paid for (energy must be spent); truth is free because it requires no energy. Observation is its invoice.

C. Glossary of Sanskrit Terms

Term	Meaning as used in this paper
Advaita Vedānta	"Non-dual" school of Vedānta; ultimate reality is non-dual consciousness
Akhaṇḍam	Undivided, without parts
Anugraha	Grace; here, the unearned availability of $w = 1$
Brahman	Ultimate non-dual reality
Cit	Consciousness
Dṛk / Dṛg	The seer (that which sees but is not seen)
Dṛśyam	The seen (any observable object)
Kāraṇa	Causal (level)
Māyā	Appearance; apparent multiplicity superimposed on non-dual reality
Neti neti	"Not this, not this"; method of negation
Samādhi	Contemplative absorption; here, the limit $w \equiv 1$
Sādhanā	Spiritual practice or discipline
Sthūla / Sūkṣma	Gross / subtle (levels)
Svayam-prakāśa	Self-luminous
Upādhi	Limiting adjunct; the filter $w(x)$
Viveka	Discrimination, discernment

Table 6: Glossary of Sanskrit terms.

References

Bergson, H. (1911). *Matter and Memory* (N. M. Paul & W. S. Palmer, Trans.). London: George Allen & Unwin. (Original work published 1896.)

Bryant, E. F. (2009). *The Yoga Sūtras of Patañjali*. New York: North Point Press.

Cover, T. M., & Thomas, J. A. (2006). *Elements of Information Theory* (2nd ed.). Hoboken, NJ: Wiley.

Deutsch, E. (1969). *Advaita Vedānta: A Philosophical Reconstruction*. Honolulu: University Press of Hawaii.

Festinger, L. (1957). *A Theory of Cognitive Dissonance*. Stanford, CA: Stanford University Press.

Friston, K. (2010). The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience*, 11(2), 127–138.

Gabor, D. (1946). Theory of communication. *Journal of the Institution of Electrical Engineers*, 93(III), 429–457.

Harris, F. J. (1978). On the use of windows for harmonic analysis with the discrete Fourier transform. *Proceedings of the IEEE*, 66(1), 51–83.

Hoffman, D. D. (2019). *The Case Against Reality: Why Evolution Hid the Truth from Our Eyes*. New York: W. W. Norton.

Huxley, A. (1954). *The Doors of Perception*. London: Chatto & Windus.

James, W. (1898). *Human Immortality: Two Supposed Objections to the Doctrine*. Boston: Houghton, Mifflin.

Kant, I. (1781/1787). *Kritik der reinen Vernunft* [*Critique of Pure Reason*]. Riga: Hartknoch.

Metzinger, T. (2003). *Being No One: The Self-Model Theory of Subjectivity*. Cambridge, MA: MIT Press.

Nikhilananda, Swami (Trans.) (1931). *Drg-Drsya-Viveka: An Inquiry into the Nature of the "Seer" and the "Seen"*. Mysore: Sri Ramakrishna Ashrama.

Nyquist, H. (1928). Certain topics in telegraph transmission theory. *Transactions of the AIEE*, 47(2), 617–644.

Potter, K. H. (Ed.) (1981). *Encyclopedia of Indian Philosophies, Vol. 3: Advaita Vedānta up to Śaṃkara and His Pupils*. Princeton, NJ: Princeton University Press.

Shannon, C. E. (1949). Communication in the presence of noise. *Proceedings of the IRE*, 37(1), 10–21.

Shannon, C. E. (1959). Coding theorems for a discrete source with a fidelity criterion. *IRE National Convention Record*, 7(4), 142–163.

Slepian, D., & Pollak, H. O. (1961). Prolate spheroidal wave functions, Fourier analysis and uncertainty I. *Bell System Technical Journal*, 40(1), 43–63.

von Neumann, J. (1932). *Mathematische Grundlagen der Quantenmechanik*. Berlin: Springer.

Zurek, W. H. (2003). Decoherence, einselection, and the quantum origins of the classical. *Reviews of Modern Physics*, 75(3), 715–775.